features. Furthermore, we can add the Spring Security Test dependency, which provides additional support for testing security scenarios in Spring.

Open the build.gradle file for the Edge Service project (edge-service) and add the new dependencies. Remember to refresh or reimport the Gradle dependencies after the new addition.

Listing 11.5 Adding dependencies for Spring Security OAuth2 Client

```
dependencies {
  ...
  implementation
  ➥ 'org.springframework.boot:spring-boot-starter-oauth2-client'
  testImplementation 'org.springframework.security:spring-security-test'
}
```

Spring integration with Keycloak

When choosing Keycloak as the Authorization Server, an alternative to the native OpenID Connect/OAuth2 support provided by Spring Security was the Keycloak Spring Adapter. It was a library supplied by the Keycloak project itself to integrate with Spring Boot and Spring Security, but it was retired after the release of Keycloak 17.

If you find yourself working on a project using the Keycloak Spring Adapter, you might want to check out my articles on the subject (www.thomasvitale.com/tag/keycloak) or chapter 9 of *Spring Microservices in Action*, second edition, by John Carnell and Illary Huaylupo Sánchez (Manning, 2021).

11.4.2 Configuring the integration between Spring Security and Keycloak

After adding the relevant dependencies on Spring Security, we need to configure the integration with Keycloak. In the previous section we